Activity 4 – Annotated Python Code

```
# Password Security Rating Program
# Evaluates password length, allowed characters, points breakdown and rating.

ALLOWED_SPECIALS = "!%&*+="
MIN_LENGTH = 8
MAX_LENGTH = 15

def rate_password(password):
    """Return result details for a password, or a validation error."""
    if not password:
        return {"error": "Password cannot be blank."}
    if len(password) < MIN_LENGTH:
        return {"error": f"Password is too short (minimum {MIN_LENGTH} characters)."}
    if len(password) > MAX_LENGTH:
        return {"error": f"Password is too long (maximum {MAX_LENGTH} characters)."}

    invalid = [char for char in password
               if not (char.islower() or char.isupper() or char.isdigit()
                       or char in ALLOWED_SPECIALS)]
    if invalid:
        return {"error": "Password contains characters that are not allowed."}

    lower_count = sum(char.islower() for char in password)
    upper_count = sum(char.isupper() for char in password)
    number_count = sum(char.isdigit() for char in password)
    special_count = sum(char in ALLOWED_SPECIALS for char in password)

    lower_points = lower_count * 5
    upper_points = upper_count * 5
    number_points = number_count * 10
    special_points = special_count * 10
    length_bonus = 20 if len(password) >= 10 else 0

    lower_penalty = lower_count * 3 if lower_count == len(password) else 0
    upper_penalty = upper_count * 3 if upper_count == len(password) else 0
    number_penalty = number_count * 5 if number_count == len(password) else 0

    total = (lower_points + upper_points + number_points + special_points
             + length_bonus - lower_penalty - upper_penalty - number_penalty)

    if total <= 20:
        rating = "Very low"
    elif total <= 40:
        rating = "Low"
    elif total <= 70:
        rating = "Medium"
    elif total <= 80:
        rating = "High"
    else:
        rating = "Very high"

    return {
        "length": len(password),
        "lower_count": lower_count, "upper_count": upper_count,
        "number_count": number_count, "special_count": special_count,
        "lower_points": lower_points, "upper_points": upper_points,
        "number_points": number_points, "special_points": special_points,
        "length_bonus": length_bonus,
        "lower_penalty": lower_penalty, "upper_penalty": upper_penalty,
        "number_penalty": number_penalty, "total": total, "rating": rating
    }

def main():
    print("PASSWORD SECURITY RATING")
    password = input("Enter a password to rate: ")
    result = rate_password(password)
    if "error" in result:
        print(result["error"])
        return

    print(f"Password length: {result['length']} characters (valid)")
    print("Points breakdown:")
    print(f"  Lowercase letters: +{result['lower_points']}")
    print(f"  Uppercase letters: +{result['upper_points']}")
    print(f"  Numeric characters: +{result['number_points']}")
    print(f"  Special characters: +{result['special_points']}")
    print(f"  Length bonus: +{result['length_bonus']}")
    print(f"  Lowercase-only deduction: -{result['lower_penalty']}")
    print(f"  Uppercase-only deduction: -{result['upper_penalty']}")
    print(f"  Numeric-only deduction: -{result['number_penalty']}")
    print(f"Total points: {result['total']}")
    print(f"Security rating: {result['rating']}")

if __name__ == "__main__":
    main()
```